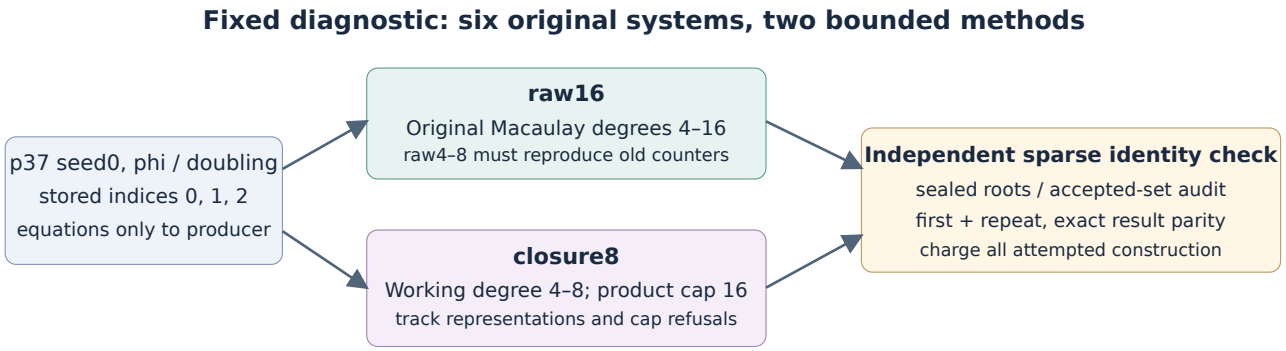
Raw degree versus degree-fall closure

Native control checkpoint, 25 October 2026. Comparison cases pending.

The fixed diagnostic supplements the full mixed-component extension-field study. It uses p37 seed0, phi and doubling, the first three stored polynomial systems per arm. Both raw16 and closure8 run on all six systems: twelve configurations, twenty-four producers, twelve first audits and twelve A/A parities. These are public synthetic systems, not independent secrets.



Observed implementation controls

Native gate v2 passes five Rust tests with zero failures. Format, tests, release build and final source-seal checks exit 0.

Control	Observed outcome
Degree fall: working degree 2; original product degree 3	Certificate expanded exactly
Nonradical system and altered multiplier	Power 2; tampering rejected
Common nonrepeated root	No bounded certificate
Insufficient original product cap	Sparse verifier rejects
Raw cap 4 versus actual product degree 1	Distinct fields, exact identity

Executable SHA-256:
`d6e4f2f6d562f0a67bd5735123d63a6d744c9e685939f4c73d9c427ad0c7a92b`

Gate parent: `6b16155b7ae747d2f00f74857b3d5b032f5a58d3`. The gate retains the then-uncommitted source snapshot. Commit the source/input freeze before any comparison execution. Input preparation and launch controls retain separate receipts; no production result is implied here.

Prelaunch review caught a raw-cap/actual-degree reporting error. V2 derives the actual product degree from multipliers and checks it separately. V1′s four-test build/report remain preserved. Only its queued infrastructure control was quiesced before admission; it is not a passing launch gate. No comparison case or original science service was stopped or repeated.

Exact boundary

raw16 reproduces the previous raw Macaulay degrees 4–8, then extends through
16. The auditor requires exact rank and arithmetic-counter equality for

the retained prefix. closure8 reduces at intermediate degrees 4–8 and multiplies newly found lower-degree rows, tracking every representation over the original equations. Its original-product cap is 16. A certificate at working degree 8 may require a larger original-product degree; report both. Count representation-cap refusals. Failure at either bound proves no nonexistence statement.

Verification, cost and remaining scope

Both methods search powers 1,2,3 of the repeated-factor polynomial $q_a = a^3 - a^2 e_1 + a e_2 - e_3$. The sufficient identity is $q_a^k = \sum_{i=0}^{3} h_i f_i$. Sparse coefficient expansion over the actual prime field checks every returned identity independently of the dense construction. The later auditor checks the sealed original roots and requires no accepted physical relation in a certified system. The producer receives no prior roots, relation outcomes or costs.

The existing conditional rejection proof and cubic IDC `IDC1h10810f17e0f6c8f6` are reused. The native parser, field, polynomial and group/decoder bridges remain open. No new general theorem is claimed.

Accounting and host

Count search and sparse-check modular multiplications, additions and inversion calls, including failed degrees. Inversion exponentiation products are already counted as multiplications. RSS, memory operations and I/O remain unpriced. No fallback, relation collection or target recovery runs in this diagnostic. Complete end-to-end cost remains null.

Linux x86-64; Intel Xeon 6975P-C; 32 logical CPUs; Rust 1.99.1 (`58cf8f9c3`, 2026-10-13). One build worker, separate Cargo target. Science phases share the busy reservation, one worker, 8 GiB address-space cap, 1,200-second outer watchdog and 1,180-second cooperative producer deadline, starting after the queue wait. Shared-host clocks are diagnostic only.

Evidence and original requirements

Source-linked report: `BUILD_AND_CONTROLS.md`; prospective specification: `PROTOCOL.md`; native source: `native/src/probe.rs` and `algebra.rs`. Raw gate: `../runs/degree-closure-probe-native-gate-v2/metadata/`. Exact public fixture SHA-256:
`5a55715c3d8f532634722ff6f58562e8f7e9f7d339f529cda724677da0f9b393`

The historical 189+21 comparison is complete and is not repeated. The full recovery and certificate panels continue under their existing owners. Unique physical relations, factor-log rank and complete cold cost remain the original comparison. These polynomial ranks are a different quantity. The new diagnostic has not yet run its twelve comparison configurations. No full-price, calibrated GAE, matched strongest IC/rho, external-tool, scale or breakthrough result is established by these controls.

Prior-art match is partial: the atlas lead `LEAD-ICE-descent-closure-degree-at-scale` and `KR-IC-fbdb61` distinguish raw spans from closure in binary descent. Their numerical bounds are not transferred to this extension-field restriction. No novelty or atlas saturation is asserted. Canonical scoreboards and benchmark index receive no row; any copied control PDF and figures stay explicitly unlisted.